

OPAL to OPALX Chicane Regression Tests

OPALX development note

June 12, 2026

1 Purpose

This note documents the consolidated chicane regression tests in `sandbox/chicane`. The directory contains two OPAL/OPALX pairs:

- `chicane-1-opal.in` and `chicane-1-opalx.in`: a five-particle hard-edge probe used to check the longitudinal dispersion of a symmetric four-SBEND chicane.
- `chicane-2-opal.in` and `chicane-2-opalx.in`: a 10000-particle manufactured-distribution comparison using the stat file quantities energy, energy spread, RMS beam sizes, RMS momenta, and RMS emittances.

The plotting script `plot_chicane_figures.py` reads the run outputs from `runs/`, writes publication-quality figures to `figures/`, and writes numerical summaries to `summaries/`. The `figures/` directory is intentionally kept figure-only.

2 Geometry

Both tests use the same idealized C-shaped chicane:

$$L_b = 1.0 \text{ m}, \quad \theta = 1 \times 10^{-1} \text{ rad}, \quad D = 1.5 \text{ m}, \quad C = 2.0 \text{ m}. \quad (1)$$

The four physical bend signs are

$$(+\theta, -\theta, -\theta, +\theta)$$

when viewed as a positive horizontal survey excursion in $+x$. For an OPAL-compatible SBEND the input length is treated as the chord length, therefore

$$\rho = \frac{L_b}{2 \sin(\theta/2)}.$$

The exit displacement of a positive survey bend is

$$\Delta x_b = \rho (1 - \cos \theta), \quad \Delta z_b = \rho \sin \theta.$$

In an OPAL `ELEMEDGE` deck the number attached to a bend is the longitudinal position of the bend entrance on the design orbit. In an OPALX explicit-pose deck, the coordinates (X, Y, Z) attached to an analytic SBEND are instead the position of the bend's local element frame. For the SBENDs used here that local frame is at the middle of the bend: it is reached by following the design orbit through half of the bend angle from the entrance. Thus the OPALX bend coordinates are not copied from the OPAL entrance `ELEMEDGE`; they are the entrance point plus the half-bend survey displacement. For a positive survey bend this displacement is

$$\Delta x_{1/2} = \rho \left(1 - \cos \frac{\theta}{2} \right), \quad \Delta z_{1/2} = \rho \sin \frac{\theta}{2}.$$

For the return bends the corresponding displacement, again written from the bend entrance point to the OPALX bend frame, is

$$\Delta x_{r,1/2} = \rho \left(\cos \frac{\theta}{2} - \cos \theta \right), \quad \Delta z_{r,1/2} = \rho \left(\sin \theta - \sin \frac{\theta}{2} \right).$$

The outer drifts are tilted by θ , so their projected increments are

$$\Delta x_D = D \sin \theta, \quad \Delta z_D = D \cos \theta.$$

3 Test 1: five-particle R_{56} probe

The first test uses five particles with

$$\delta = \frac{pz}{p_0} - 1 \in \{-2, -1, 0, 1, 2\} \times 10^{-3}, \quad p_0 = 1957.950928190185.$$

The plotting script reads the final monitor H5 files and fits final local z against the initial δ . The transport coefficient is reported as

$$R_{56} = -\frac{dz_{\text{exit}}}{d\delta}.$$

The small-angle hard-edge reference used for this chicane is

$$R_{56,\text{ref}} \simeq -\theta^2 \left(\frac{4L_b}{3} + 2D \right) = -4.33333333333334 \times 10^{-2} \text{ m}.$$

The current generated summary gives

$$R_{56,\text{OPALX}} = -4.334628766121465 \times 10^{-2} \text{ m}, \quad R_{56,\text{OPAL}} = -4.337546378253016 \times 10^{-2} \text{ m}.$$

4 Test 2: 10000-particle stat comparison

The second test uses a deterministic manufactured distribution with 10^4 particles, a 5 ps time step, and a nonzero bunch charge. The stat-file comparison deliberately ignores monitor files and compares only quantities available in the OPAL and OPALX stat files. OPAL samples are interpolated onto the OPALX s -grid over the shared longitudinal interval. For every quantity q , the lower-panel relative difference is

$$|\Delta q|[\%] = 100 \frac{|q_{\text{OPALX}}(s) - q_{\text{OPAL}}(s)|}{|q_{\text{OPAL}}(s)|}.$$

Values with a denominator at numerical zero are masked.

5 Moving an OPAL input deck to OPALX

The chicane examples expose two different migration modes. The first mode is compatibility placement: keep the old OPAL reference-line `ELEMEDGE` description and ask OPALX to interpret the deck as close as possible to OPAL. The second mode is native OPALX placement: compute explicit element poses $(X, Y, Z, \Theta, \Phi, \Psi)$ and place each element body in the global survey.

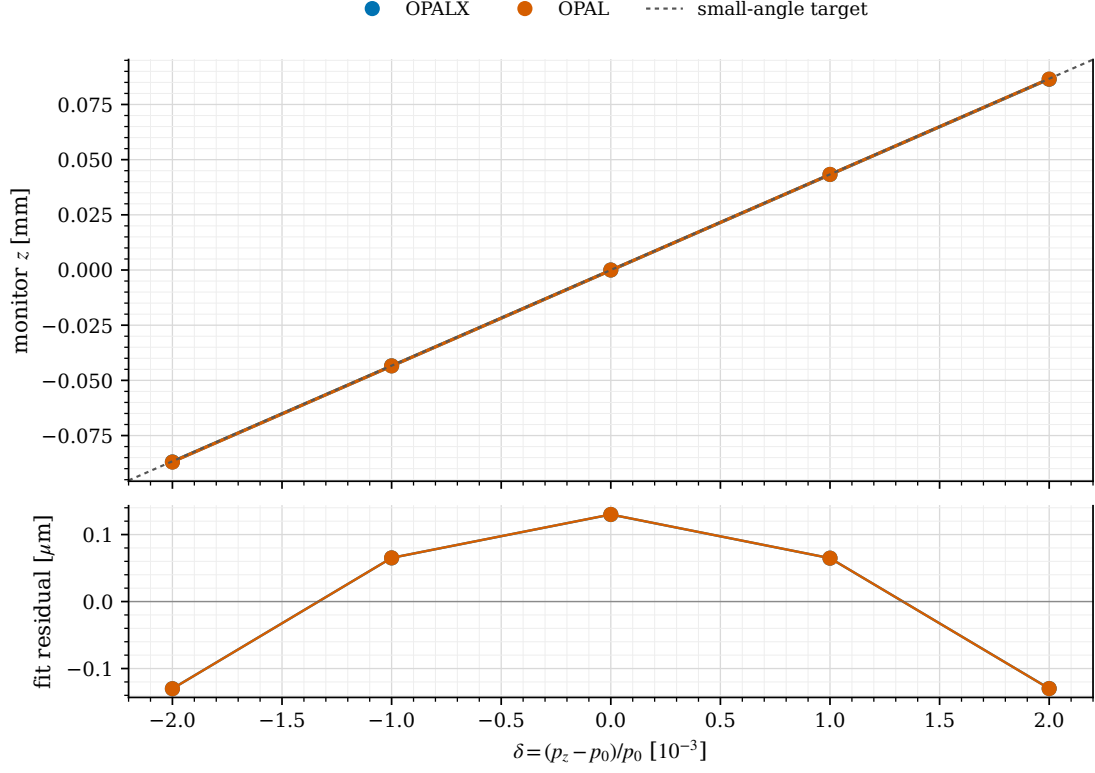


Figure 1: Five-particle R_{56} probe. The top panel shows the fitted final monitor coordinate versus the input momentum offset. The lower panel shows the fit residual.

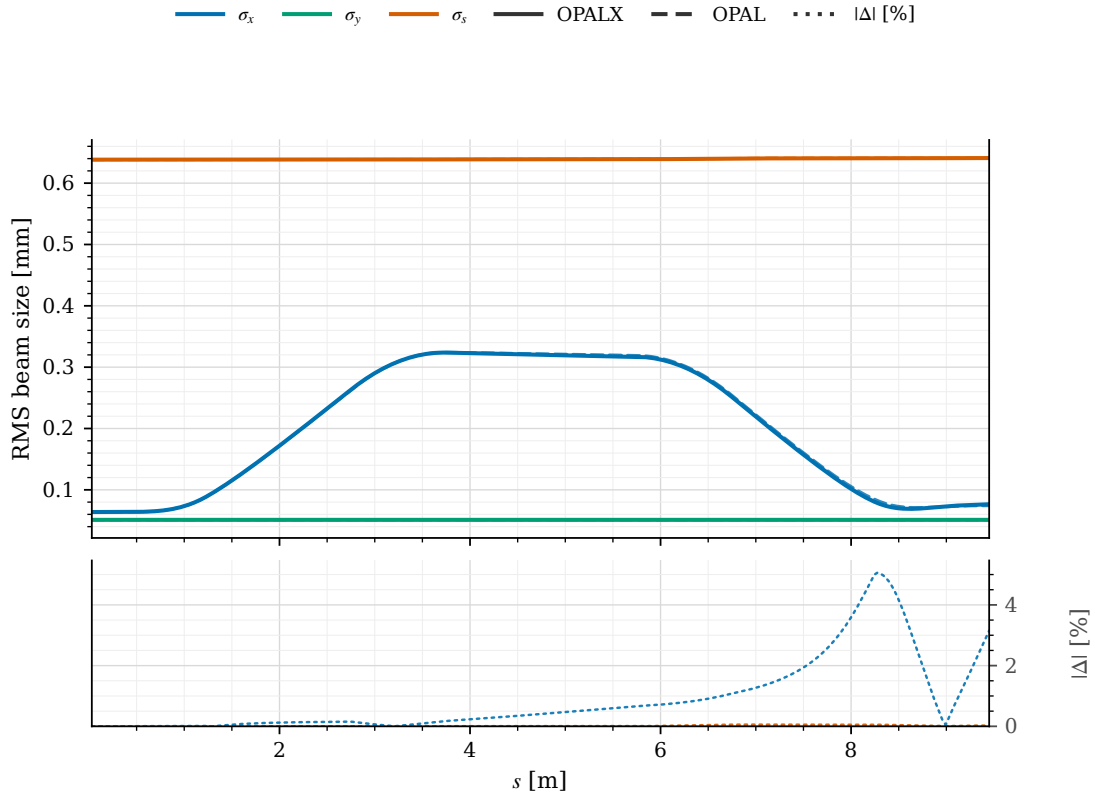


Figure 2: RMS beam-size comparison from the stat files.

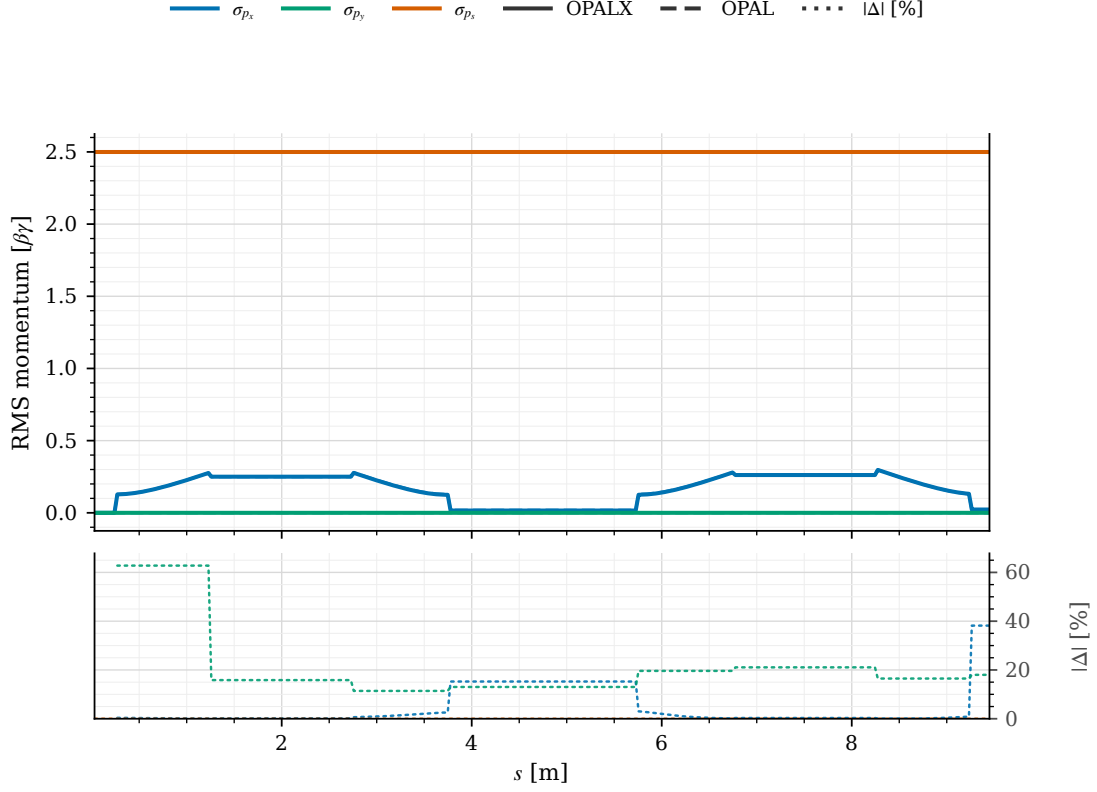


Figure 3: RMS momentum comparison from the stat files.

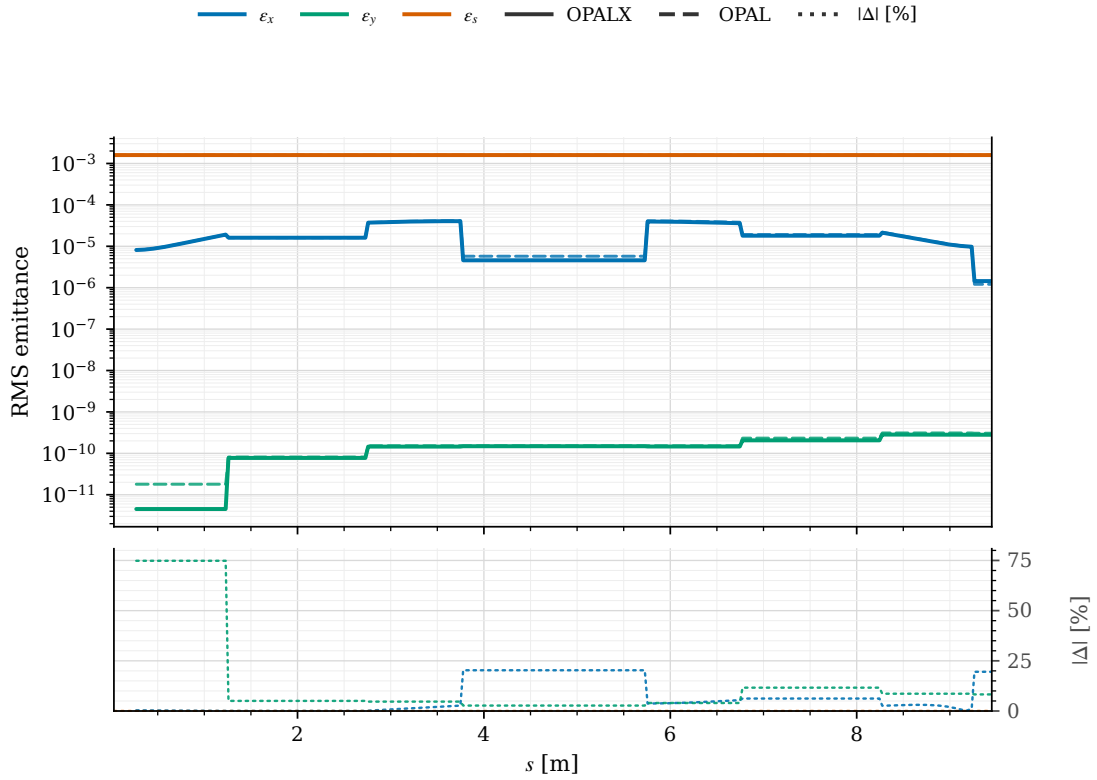


Figure 4: RMS emittance comparison from the stat files. The emittance axis is logarithmic because the transverse and longitudinal emittances differ by many orders of magnitude.

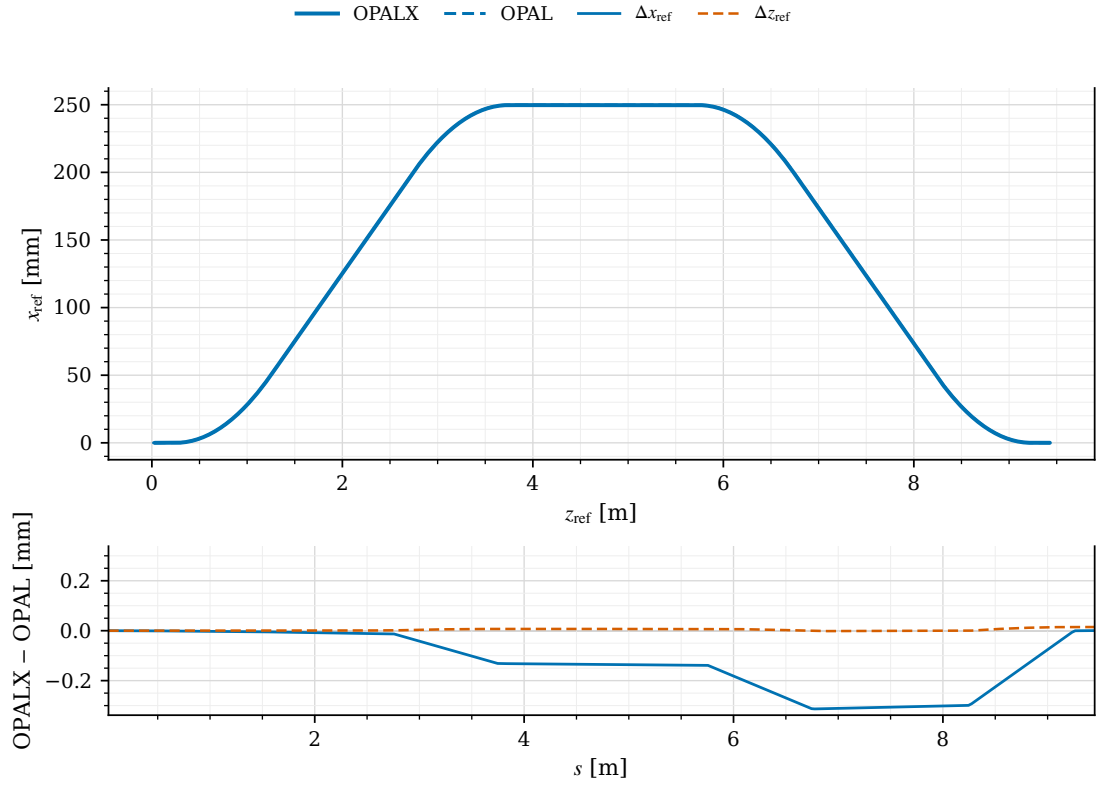


Figure 5: Reference-orbit comparison and OPALX-minus-OPAL reference-orbit difference.

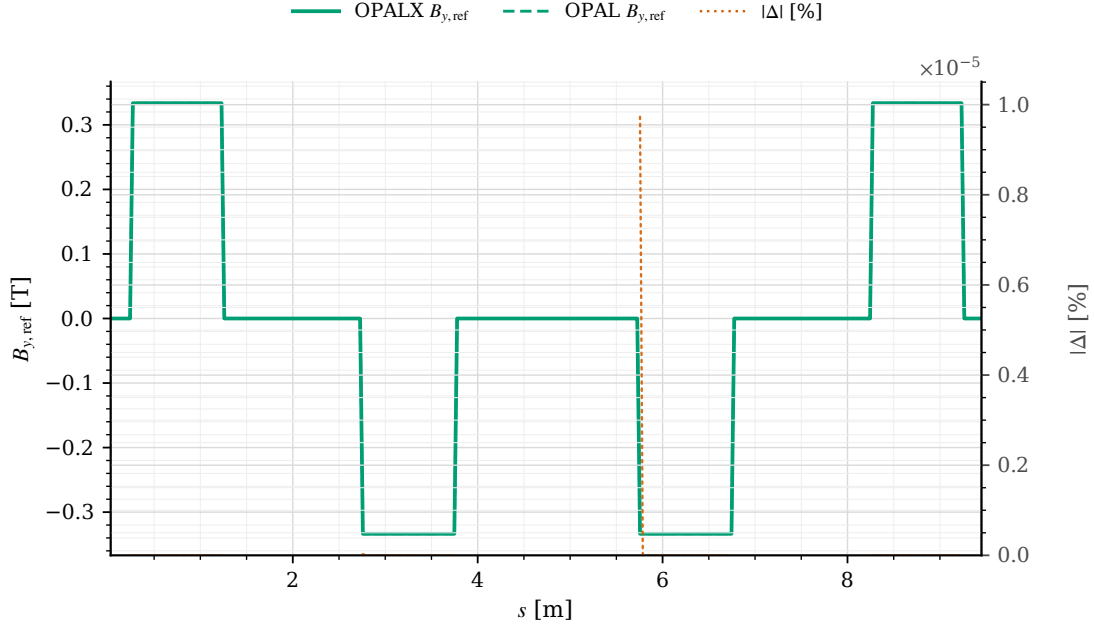


Figure 6: Reference-particle B_y comparison. This plot is diagnostic: spikes at element interfaces are useful for checking placement and temporal crossing behavior, but the main beam comparison is based on stat-file RMS quantities.

5.1 Compatibility placement

Use this mode when the goal is to reproduce an existing OPAL deck before changing the lattice description. The chicane-2 OPALX deck follows this approach.

1. Keep the OPAL element order and reference-line lengths. For the chicane this is

$$(DPRE, B1, D12, B2, D23, B3, D34, B4, MEXIT).$$

2. Keep ELEMEDGE values from OPAL:

$$\begin{aligned} B1 : & D_{\text{pre}}, \\ B2 : & D_{\text{pre}} + L_b + D, \\ B3 : & D_{\text{pre}} + 2L_b + D + C, \\ B4 : & D_{\text{pre}} + 3L_b + 2D + C. \end{aligned}$$

3. Keep OPAL bend model parameters that affect the old analytic field: FMAPFN, DESIGNENERGY, FINT, GAP, and HAPERT. In this hard-edge comparison FINT=0 suppresses fringe focusing.
4. Add OPTION, IDEALIZED = TRUE; when the purpose is an OPAL compatibility check. This keeps the nominal design orbit on hard-edge reference-line lengths instead of applying profile-dependent path corrections during compatibility placement. The option is described in more detail below.
5. Translate the OPAL fieldsolver names: FSTYPE=NONE becomes TYPE=NONE; MX,MY,MT become NX,NY,NZ; PARFFT becomes PARFFTZ; and BCFFT becomes BCFFTZ.
6. Translate the distribution and beam setup. OPAL uses DISTRIBUTION directly in RUN. OPALX uses an EMISSIONSOURCE, an EMISSIONSOURCELIST, and a BEAM with SOURCES. For a fixed from-file distribution, set NPARTDIST on the distribution and NALLOC on the beam.
7. Translate the run method. OPAL METHOD="PARALLEL-T" becomes OPALX METHOD="PARALLEL".

5.2 Meaning of IDEALIZED=TRUE

IDEALIZED is a global OPALX option used by the compatibility placement compiler. In the current code path it affects bends that do not already have an explicit body pose. That is, it acts on OPAL-style elements placed by ELEMEDGE; it does not move elements whose placement is already fixed by $(X, Y, Z, \Theta, \Phi, \Psi)$.

The compatibility-placement algorithm constructs a design orbit from the OPAL-style sequence of ELEMEDGE anchors. This is the same accelerator language used in introductory beam-dynamics treatments: particle coordinates are described in a local frame attached to the design orbit followed by the design particle. For an element whose OPAL-style entrance coordinate is

$$s_i = \text{ELEMEDGE}_i,$$

the straight drift from the previous element end to this entrance is taken as

$$\Delta s_i = s_i - s_{\text{design, end}, i-1}.$$

After placing a bend, OPALX must choose the design-orbit path length used to advance the reference-line end position to the next element. Denote this length by ℓ_{place} . The design-orbit update is

$$s_{\text{design, end}, i} = s_i + \ell_{\text{place}}.$$

With `IDEALIZED=TRUE`, OPALX uses the hard-edge design length for this design-orbit advance:

$$\ell_{\text{place}} = \ell_{\text{hard}}.$$

For an ideal sector bend this is the usual arc length

$$\ell_{\text{hard}} = \rho|\theta|,$$

while the body displacement used for the 3D placement is still the chord geometry

$$\mathbf{c} = L_{\text{chord}} R_{\text{axis}}\left(\frac{\theta}{2}\right) \hat{\mathbf{z}}.$$

For the chicane compatibility deck this means the next `ELEMEDGE` anchor is interpreted exactly on the hard-edge OPAL reference line. This is the intended mode for reproducing an old OPAL input before introducing native OPALX body poses.

With `IDEALIZED=FALSE`, OPALX replaces the hard-edge design-orbit advance by a profile-dependent length derived from the bend design path. In the code this is

$$\ell_{\text{place}} = \ell_{\text{true}} - d_{\text{entry}} - d_{\text{exit}},$$

where ℓ_{true} is the path length of the computed design path over the field support, and

$$d_{\text{entry}} = \|\mathbf{r}_{\text{true}}(s_{\text{min}})\|,$$

$$d_{\text{exit}} = \|R_z \mathbf{r}_{\text{true}}(s_{\text{max}}) - \mathbf{r}_{\text{exit,HE}}\|.$$

Here $\mathbf{r}_{\text{true}}$ is the bend design path in the local bend chart, R_z is the configured bend rotation about the global z -axis, and $\mathbf{r}_{\text{exit,HE}}$ is the hard-edge exit point. In the implementation the hard-edge exit vector is constructed as

$$\mathbf{r}_{\text{exit,HE}} = L_{\text{chord}} R_{\text{axis}}\left(\frac{\theta}{2} - E_1\right) \hat{\mathbf{z}},$$

with bend angle θ , entrance edge angle E_1 , chord length L_{chord} , and effective bend axis axis = $R_z(0, -1, 0)$.

The practical consequence is that `IDEALIZED=FALSE` can shift the downstream design-orbit end position when a bend has finite fringe-field support or another non-hard-edge design-path correction. That is useful when the OPALX lattice should follow the effective field-supported design path. It is not the right setting for a strict OPAL `ELEMEDGE` compatibility test, because old OPAL decks normally place the next element by the hard-edge reference length written in the input.

5.3 Native OPALX body-pose placement

Use native placement when the goal is to exploit OPALX's more general lattice model or to make the survey explicit. This is the mode used by `chicane-1-opalx.in`.

1. Choose the entrance survey origin. In the chicane-1 deck the entrance is $(X_0, Y_0, Z_0) = (0, 0, 0)$ with tangent along $+z$.
2. Convert every old `ELEMEDGE` into a physical survey position. Drifts and monitors are placed at their entrance or plane position. For these analytic SBENDs, OPALX **x/y/z** denotes the bend's local element frame at the middle of the bend, not the entrance point. Therefore the half-bend survey displacement above must be added to the entrance point obtained from the OPAL `ELEMEDGE`.

3. Set the bend orientation explicitly. For the positive survey convention used here, B1 and B4 have body angles $\Theta = \pm\theta/2$ and B2/B3 are oriented to close the chicane according to the local tangent. The monitor planes use the tangent angle at their location.
4. Check the OPALX SBEND sign convention. In the native body-pose chicane, the OPALX ANGLE sign is opposite to the positive-X survey convention: B1 and B4 use ANGLE=-TH; B2 and B3 use ANGLE=TH.
5. Replace OPAL aperture/fringe fields with OPALX-native hard-edge settings when appropriate. For this hard-edge regression, HGAP=0 and FINT=0 are used in the native OPALX deck.
6. Place diagnostic monitors explicitly using X/Y/Z and THETA/PHI/PSI. For the R_{56} test this is essential, because the final local monitor coordinate is fitted against the input momentum offset.

5.4 Quantities that need special attention

- **Longitudinal coordinate:** OPAL and OPALX do not necessarily write monitor crossings with identical temporal sampling. For the 10000-particle comparison the robust quantity is therefore the stat file sampled on the shared s -range.
- **Bend placement:** if OPAL and OPALX disagree at bend interfaces, inspect the reference-particle field plot and the reference-orbit difference before changing physics parameters.
- **Sampling:** STATDUMPFREQ controls the density of the stat comparison. Values of 10 or 20 are useful for chicane-2; denser output is usually only needed for debugging.
- **Relative differences:** use the same denominator everywhere: OPAL interpolated onto the OPALX s -grid. This avoids mixing sampling artifacts with physics differences.

6 Reproducing the figures

From the repository root, regenerate all figures and summaries with

```
PYTHONPYCACHEPREFIX=/private/tmp/opalx-pycache \
MPLCONFIGDIR=/private/tmp/opalx-matplotlib \
/Users/adelman/.venv-h6/bin/python sandbox/chicane/plot_chicane_figures.py
```

The script expects the run outputs to already exist under `sandbox/chicane/runs`. It does not rerun OPAL or OPALX.